铣刀的切削参数(2)
Explanation on Milling Conditions (2)

※备注 Remarks

- **30° 螺旋角 Helix 30°**

  可支持沟槽和侧面加工的通用型。
  Recommendable for both slotting and side milling.

- **35° 螺旋角、40° 螺旋角、45° 螺旋角 Helix 35"°, Helix 40°, Helix 45°**

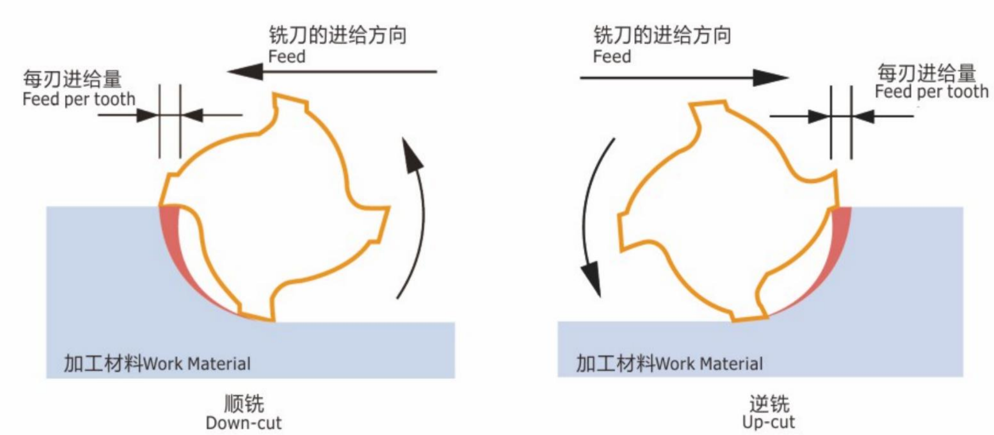
  根据加工表面所需的倾斜程度和起伏量进行选择。
  螺旋角越大，锋利度越好，振刀也越少，还能加大刀刃的长度。
  Selectable within the required tolerance.
  As larger angle gives higher shearing ability and reduce chattering, it is possible to relatively prolong the length of cut.

- **45° 螺旋角 Helix 45°**

  对淬火钢及难削材的加工有效。
  Suitable for machining for hardened steels and tough materials.

切削方向(逆铣与顺铣) Cuting Direction (Upcut and Down-cut)

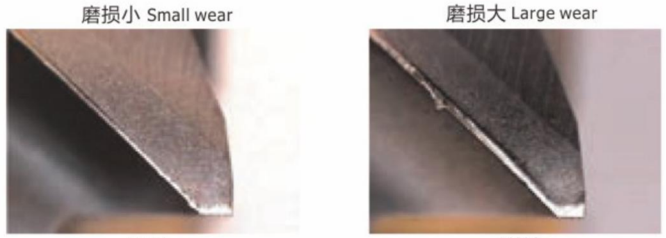
逆铣与顺铣的切削机构图 Cutting structure of Up-cut and Down-cut



不同切削方向的刀具磨损量比较图片 Diference of edge wear by utting diretions

(对S50C进行50m切削加工后的状态After cutting of 50m, S50C)

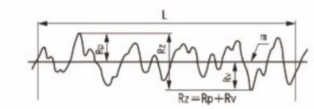
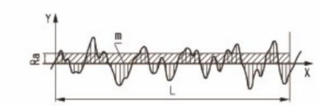
磨损小 Small wear　　磨损大 Large wear



顺铣 Down-cut　　逆铣 Up-cut

顺铣是从厚进刀缓缓切薄,若是逆铣则按照相反的方向进行切削加工。从图片上也可得知，一般情况下，顺铣时刀具的损耗少，使用寿命长。
Down-cut tooth first cuts thicker then progressively thinner, while Up-cut goes the opposite. As shown by above photos, Down-cut is recommended since the wear of cutting edge is comparatively small and tool life is eventually longer.

各种表面粗糙度的计算方法(JIS B 0601-2001)
Measurement and Symbol of Surface Roughness (Extrac from JIS B 0601 -2001)

种类 Type	符号 Symbol	计算方法 Description	说明图 Reference chart
最高值 Maximum Height of the Profile Profile	Rz	在粗糙度曲线上，沿着平均线方向截取一段基准长度，然后在所截取的基准长度范围内，沿着粗糙度曲线的纵向倍率方向测量从峰顶到谷底的间隔。该间隔的值就是粗糙度的最高值，单位为μm。截取基准长度时，如果峰顶及谷底的值大得过于极端，则可视为伤痕,不能截取。 The maximum height of the profile is the distancebetween the maximum peak height and the maximum valley depth from the mean line in each sampling length. Rz (SO/JIS) is the mean value of the maximum peak-to-valley heights in the evaluation length.	
平均值 Arithmetic Mean Deviation of the	Ra	在粗糙度曲线上,沿着平均线方向截取一段基准长度，然后将所截取部分的平均线方向定为x轴，将纵向倍率方向定为y轴，当粗糙度曲线以y=f(X)表示时,按照右图中所述的换算公式所求得的值就是平均值，单位为μm。 The arithmetic mean deviation of the profile is the arithmetic mean of the absolute values of distances from the mean line to the profile.	 $Ra=\frac{1}{L}\int_0^L \lvert f(x)\rvert dx$ L =基准长度 Evaluation Length

最高值Rz的区分值 Value of Maximum Height of the Profile	平均值Ra的区分值 Value of Arithmetic Mean Deviation of the Profile	传统的三角符号 Finishing Symbol	表面特征的属性(Ra示例时) Indication of surface texture in technical product documentation
0.05S 0.1S 0.2S 0.3S 0.5S	0.012a 0.025a 0.05a 0.1a 0.2a	▼▼▼▼	√Ra 0.012 ~ √Ra 0.2
1.6S 3.2S 6.3S	0.4a 0.8a 1.6a	▼▼▼	√Ra 0.4 ~ √Ra 1.6
12.5S 25S	3.2a 6.3a	▼▼	√Ra 3.2 ~ √Ra 6.2
50S 100S	12.5a 25a	▼	√Ra 12.5 ~ √Ra 25

※Ra、Rz与三角符号的相互关系，只是为了方便参考，并无严密性。
※Note: Triangle Finishing Symbol Mark presents approximate surface roughness specified by Ra and Rz.

技术资料 Technical Guidance